

Pairwise F-test Results

2024 transfer-function polynomial scan

Central and forward categories

Candidates: 0x0, 0x1, 0x2, 1x0, 1x1, 1x2, 2x1, 2x2

CMS Work in Progress

13.6 TeV

Comparison

Each panel compares a simpler polynomial order with a more complex one.

Observed F

The blue arrow marks the F statistic calculated from the two goodness-of-fit results.

p-value

A small p-value indicates that the additional parameters improve the fit beyond the simpler model.

Important

These are polynomial-order comparisons within the same transfer-function construction, not tests of unrelated functional families.

Physics interpretation and the final polynomial choice should also consider fit stability and diagnostics.

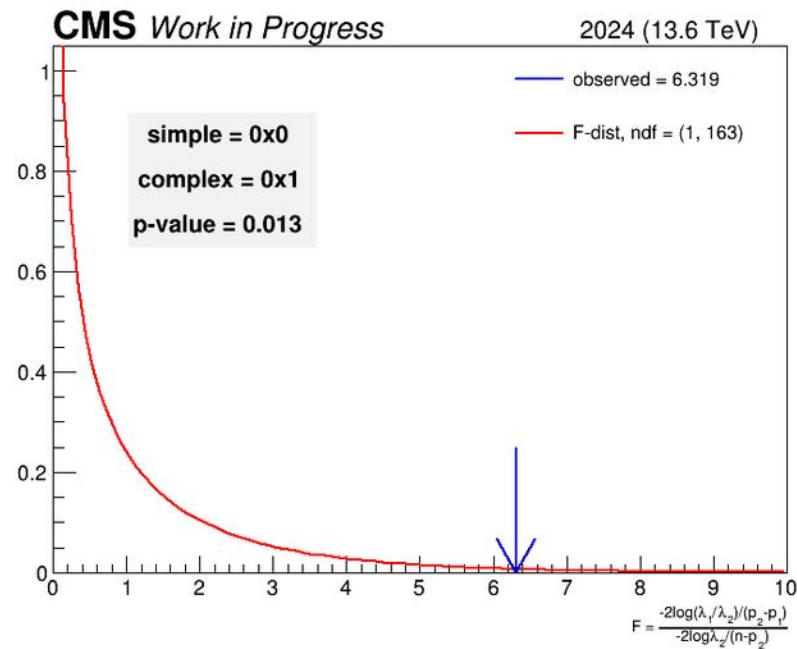
Central category

25 pairwise comparisons

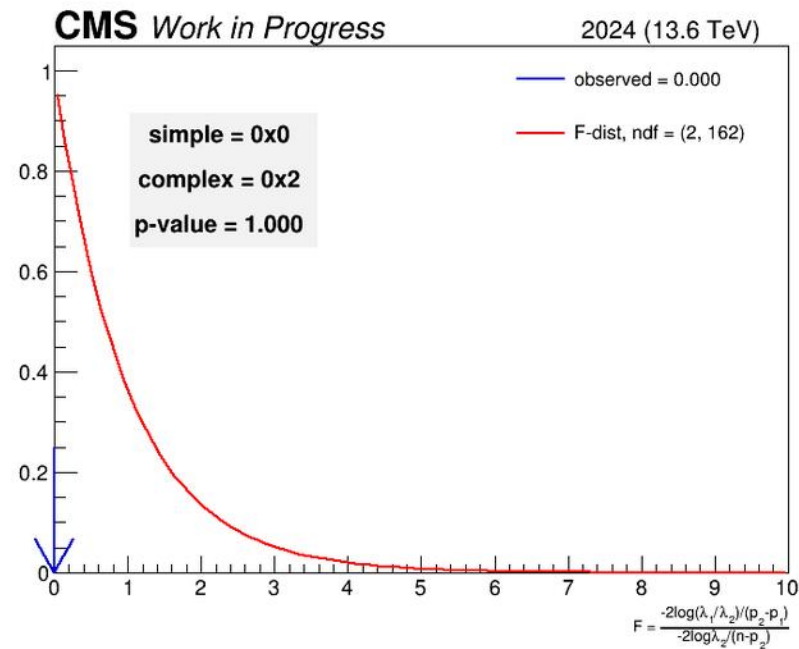
11 comparisons have $p < 0.05$

Counts summarize pairwise tests only; they do not define the final model choice.

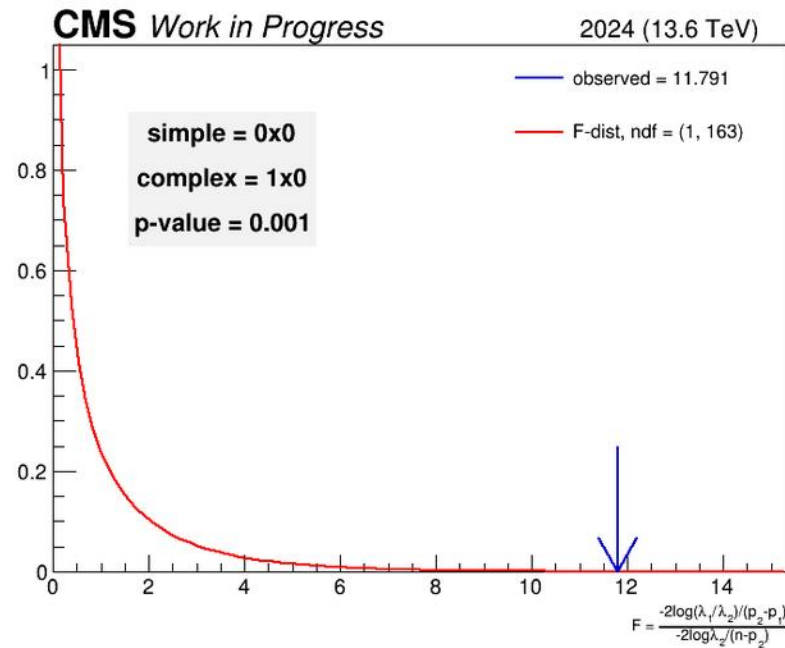
0x0 vs 0x1 | $F = 6.32$ | $p = 0.0129$



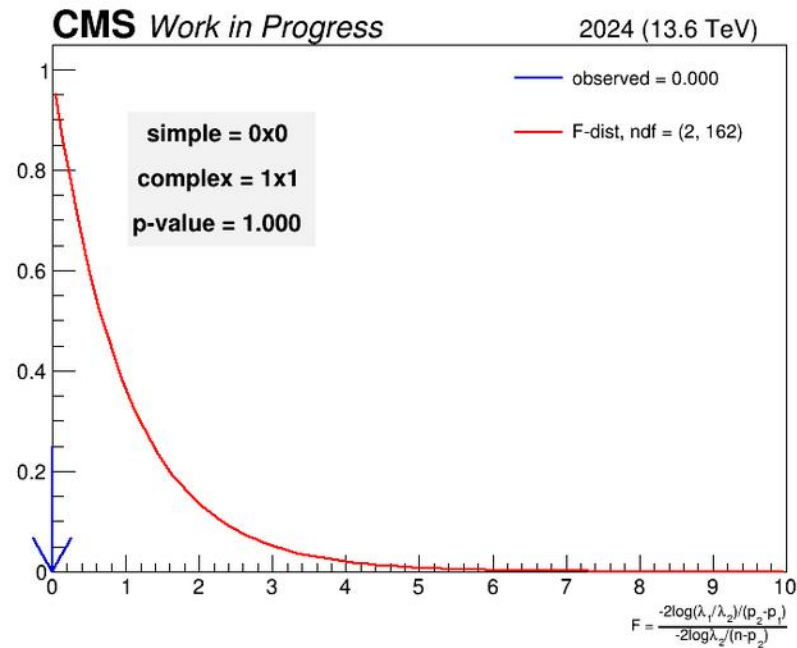
0x0 vs 0x2 | $F = 0$ | $p = 1$



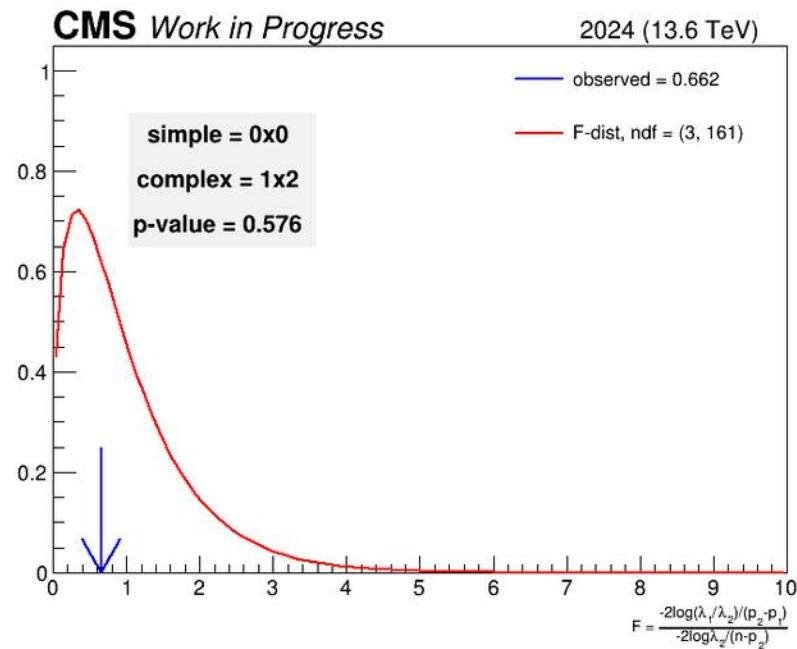
0x0 vs 1x0 | $F = 11.8$ | $p = 0.000755$



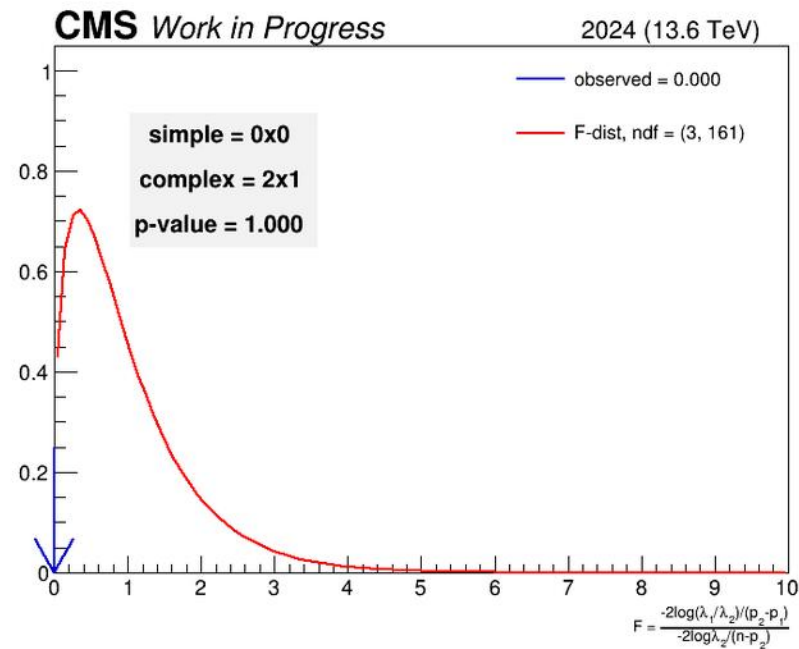
0x0 vs 1x1 | $F = 0$ | $p = 1$



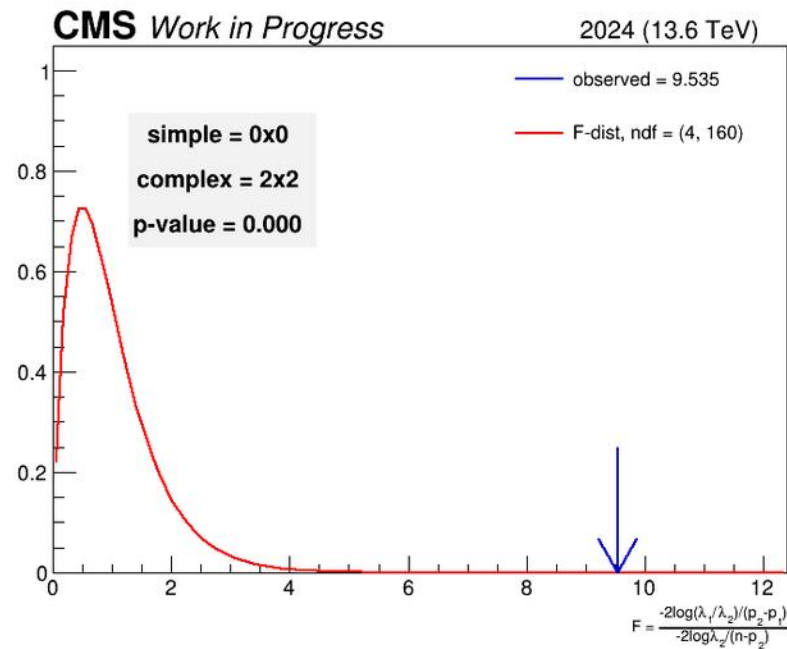
0x0 vs 1x2 | F = 0.662 | p = 0.576



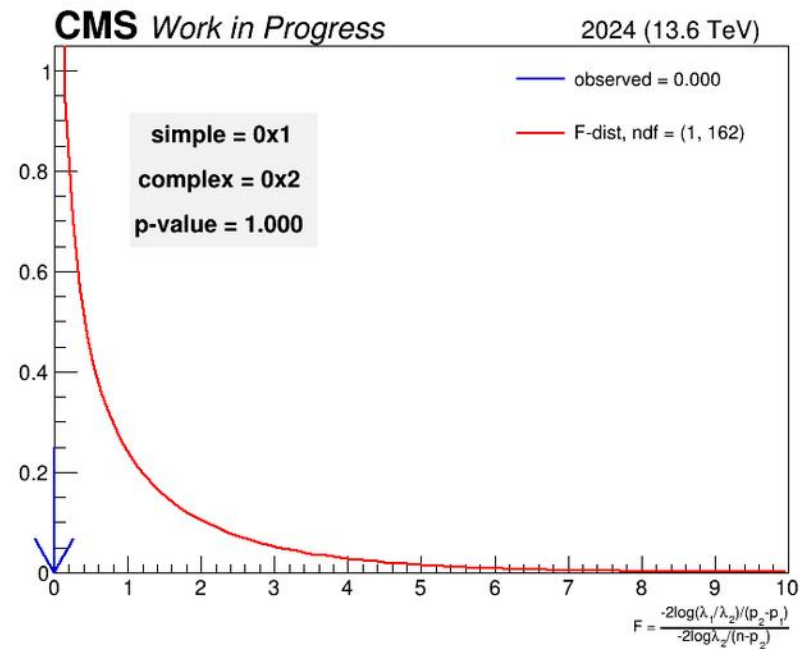
0x0 vs 2x1 | F = 0 | p = 1



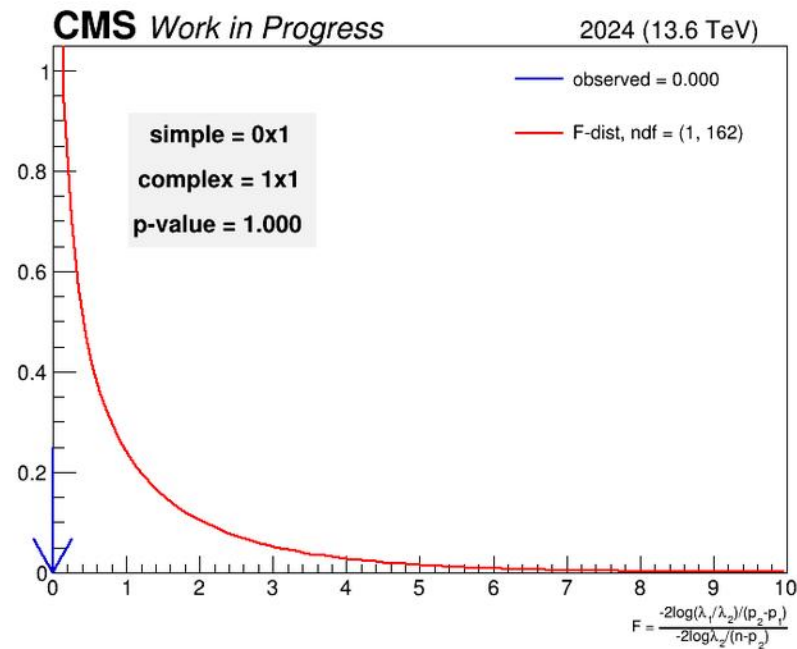
0x0 vs 2x2 | $F = 9.53$ | $p = 6.12\text{e-}07$



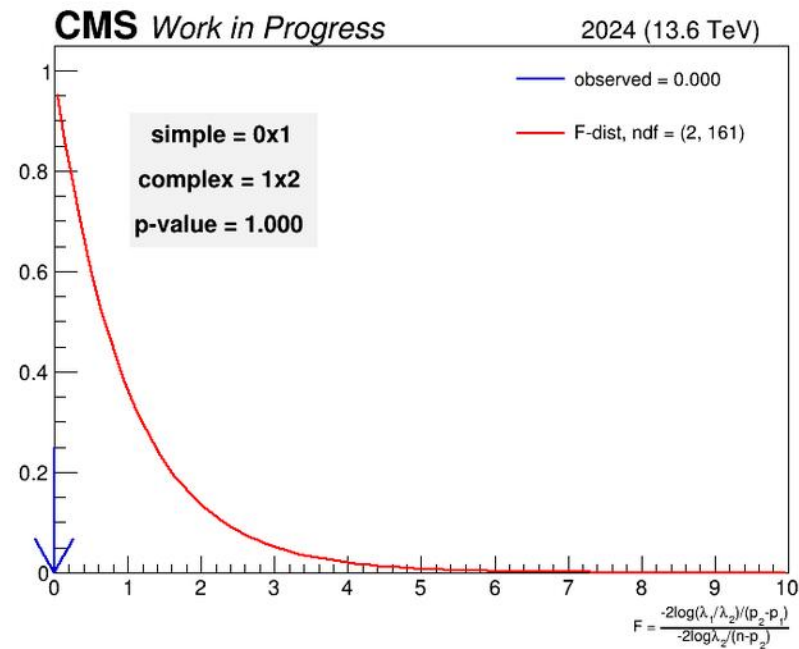
0x1 vs 0x2 | $F = 0$ | $p = 1$



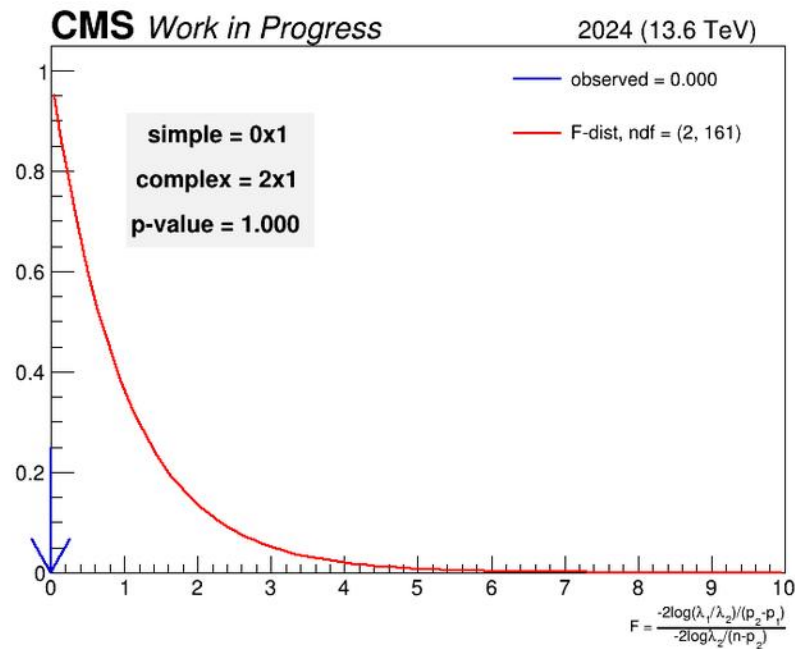
0x1 vs 1x1 | $F = 0$ | $p = 1$



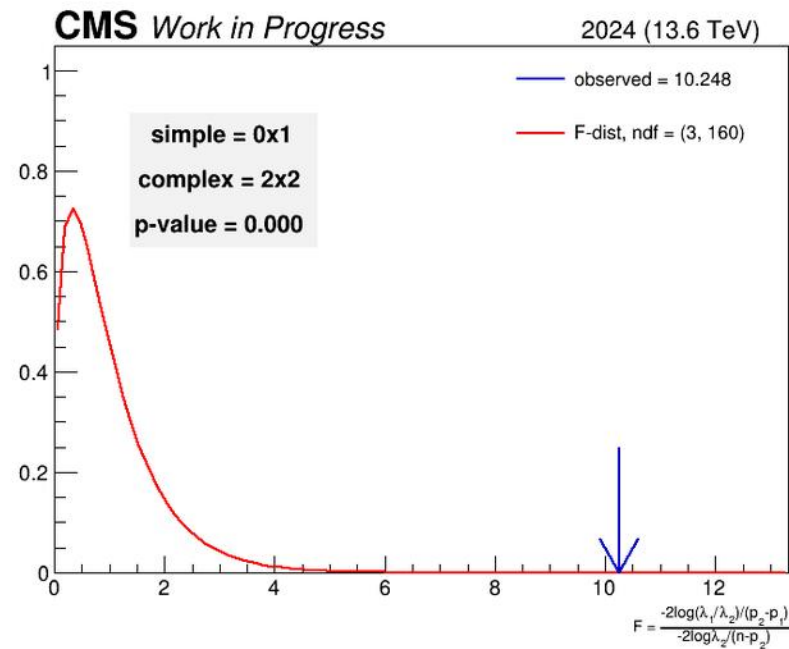
0x1 vs 1x2 | $F = 0$ | $p = 1$



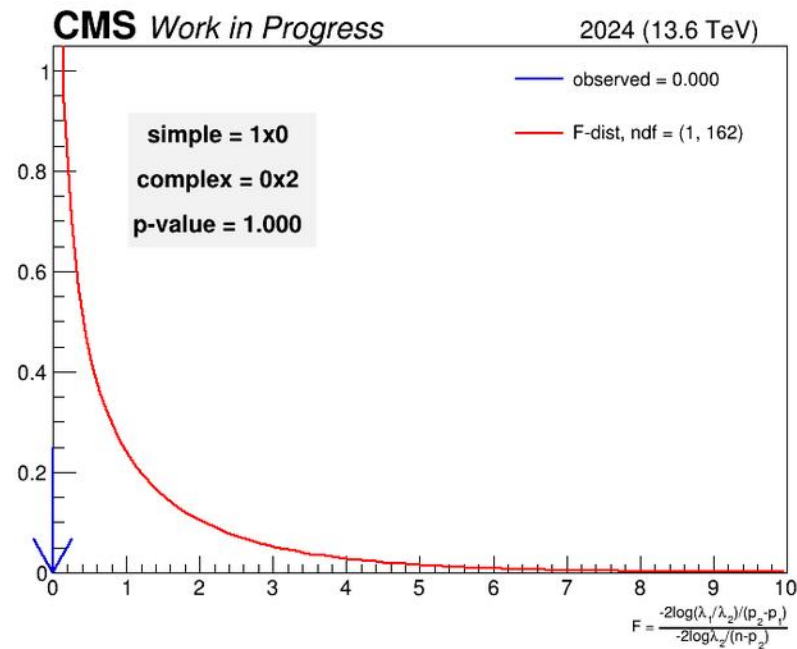
0x1 vs 2x1 | $F = 0$ | $p = 1$



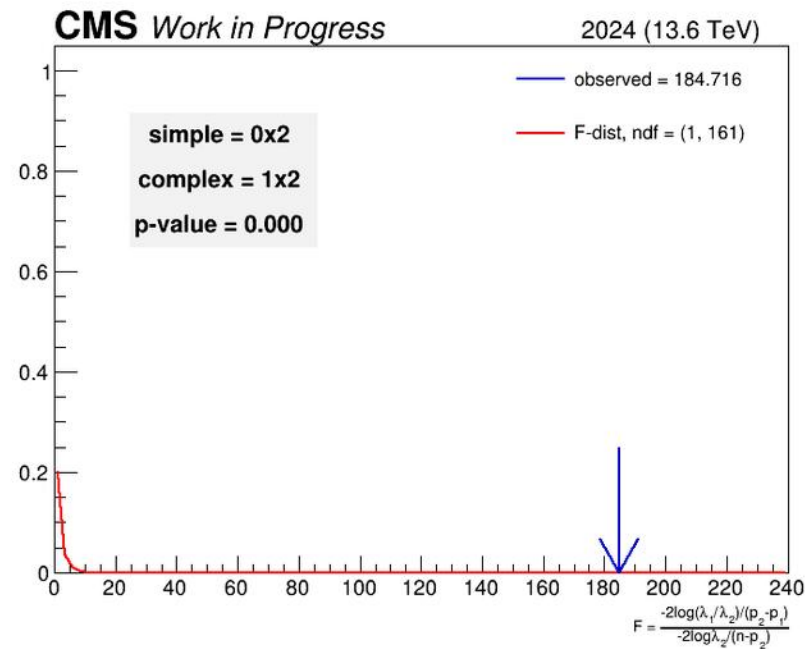
0x1 vs 2x2 | $F = 10.2$ | $p = 3.28\text{e-}06$



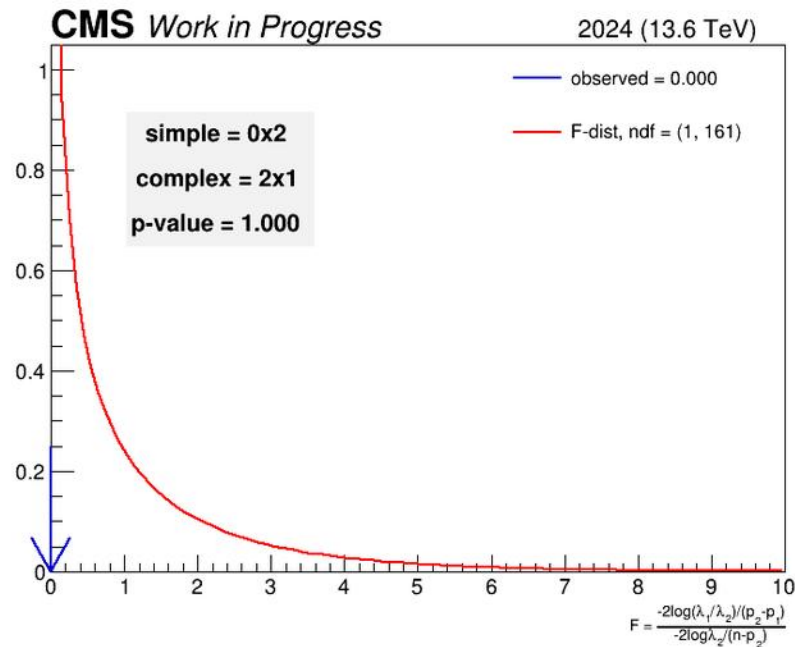
1x0 vs 0x2 | $F = 0$ | $p = 1$



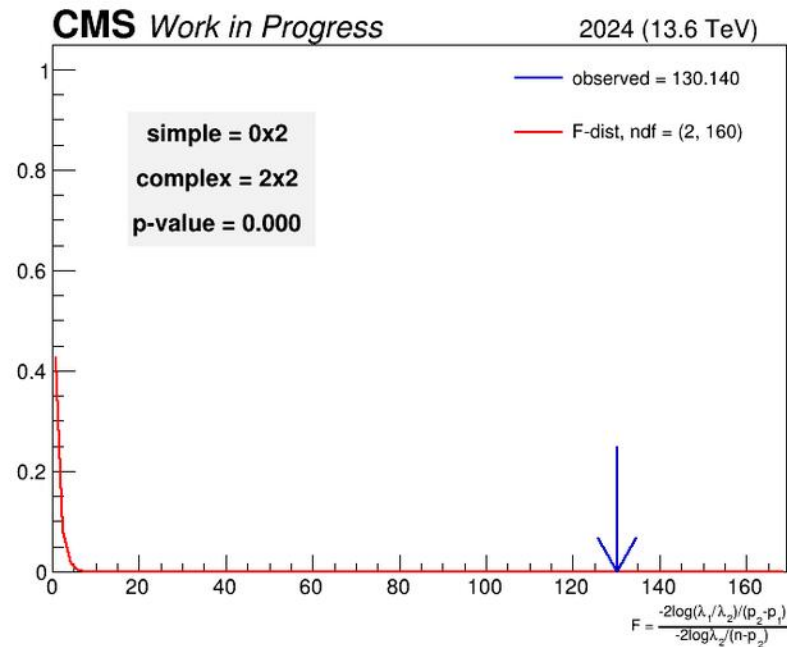
0x2 vs 1x2 | $F = 185$ | $p = < \text{numerical precision}$



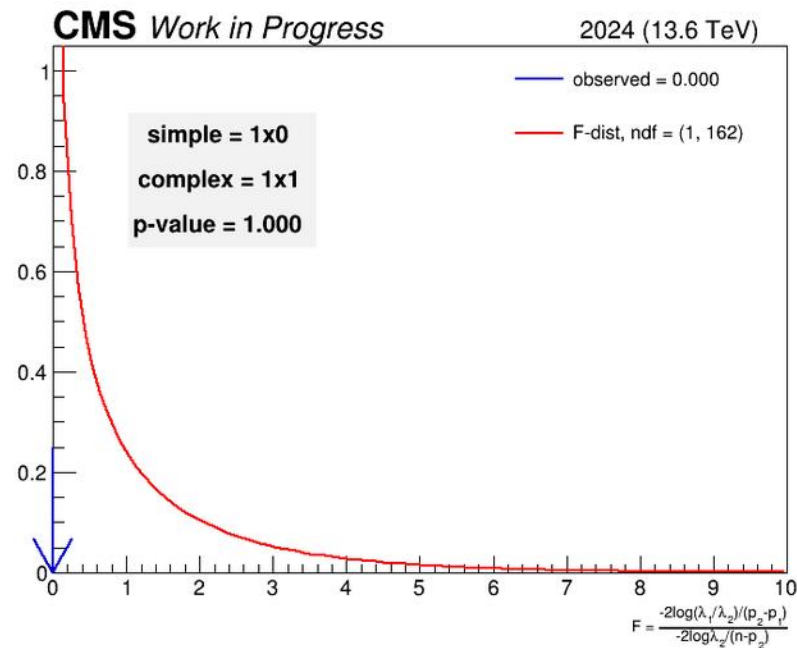
0x2 vs 2x1 | $F = 0$ | $p = 1$



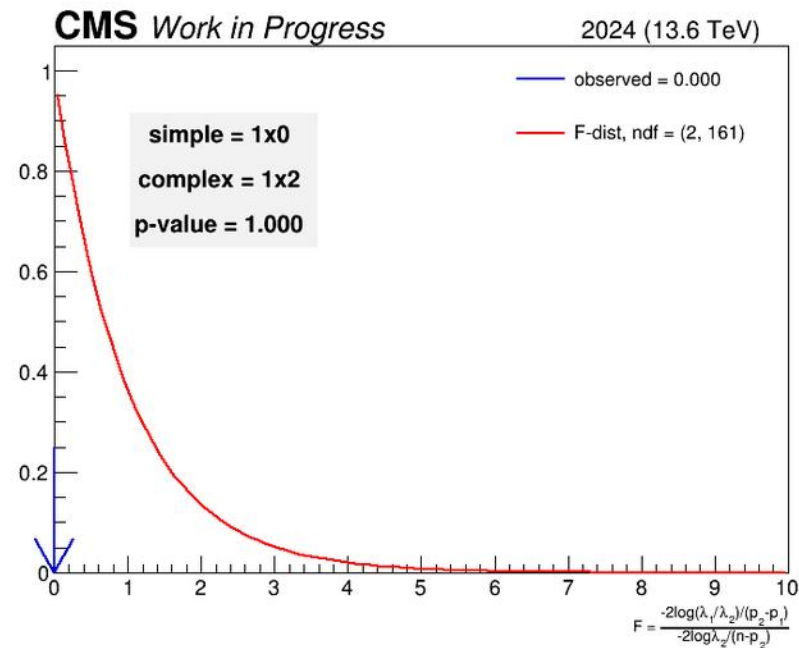
0x2 vs 2x2 | $F = 130$ | $p = < \text{numerical precision}$



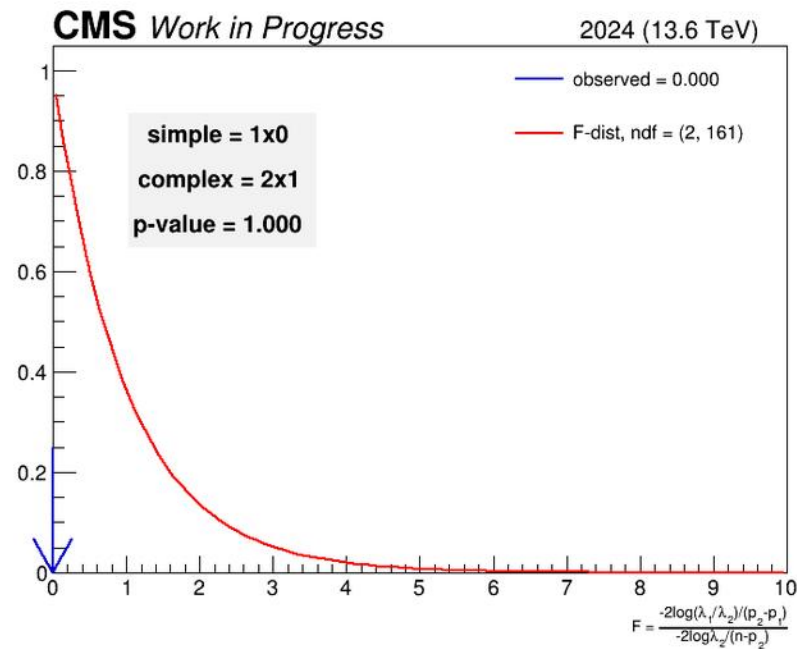
1x0 vs 1x1 | $F = 0$ | $p = 1$



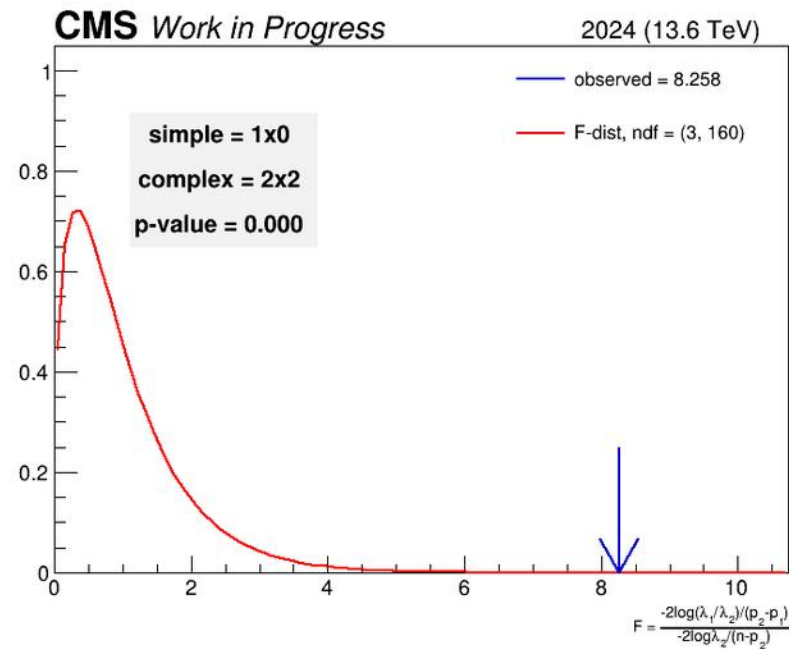
1x0 vs 1x2 | $F = 0$ | $p = 1$



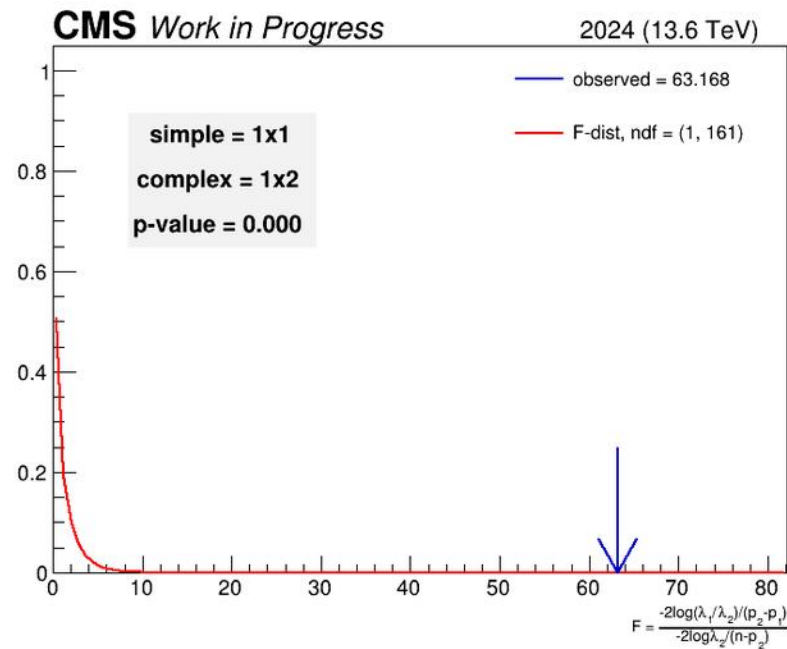
1x0 vs 2x1 | F = 0 | p = 1



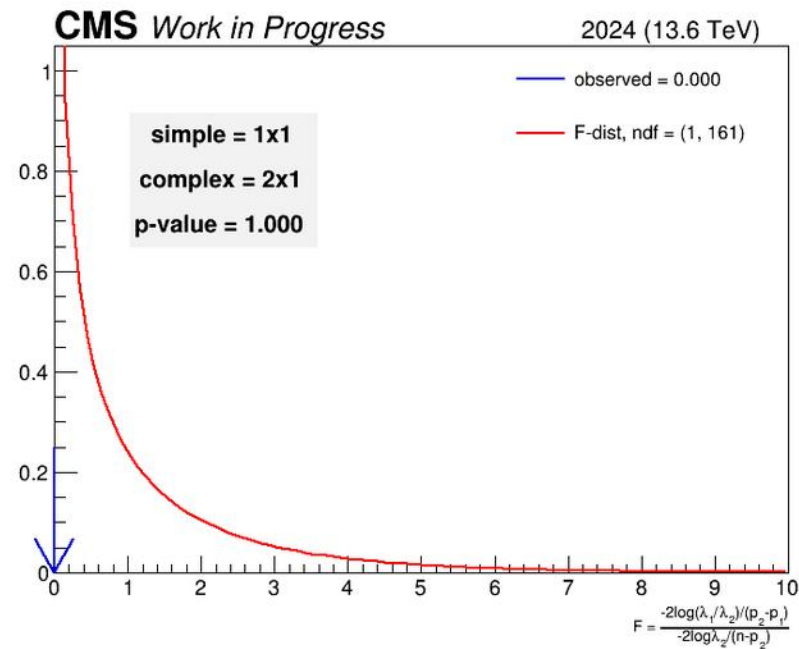
1x0 vs 2x2 | F = 8.26 | p = 3.84e-05



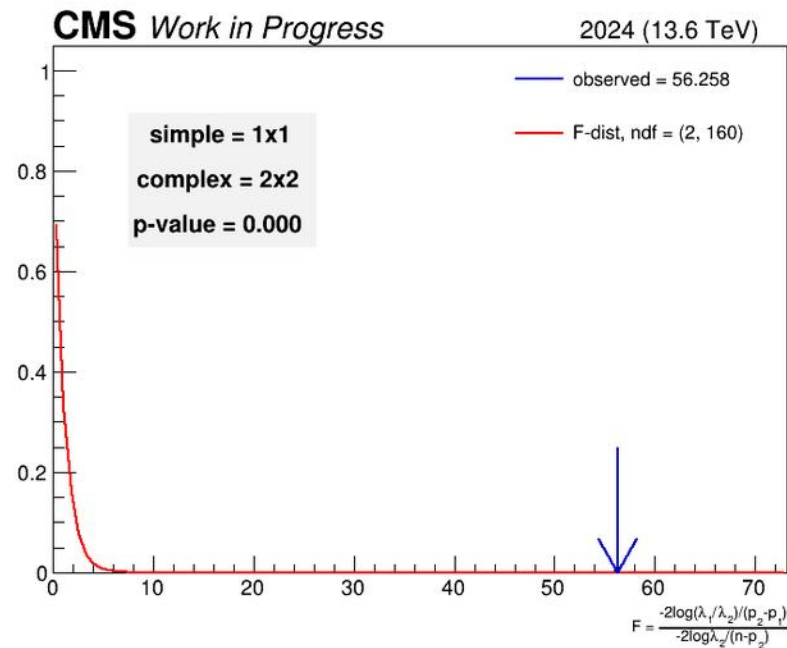
1x1 vs 1x2 | $F = 63.2$ | $p = 5.03e-13$



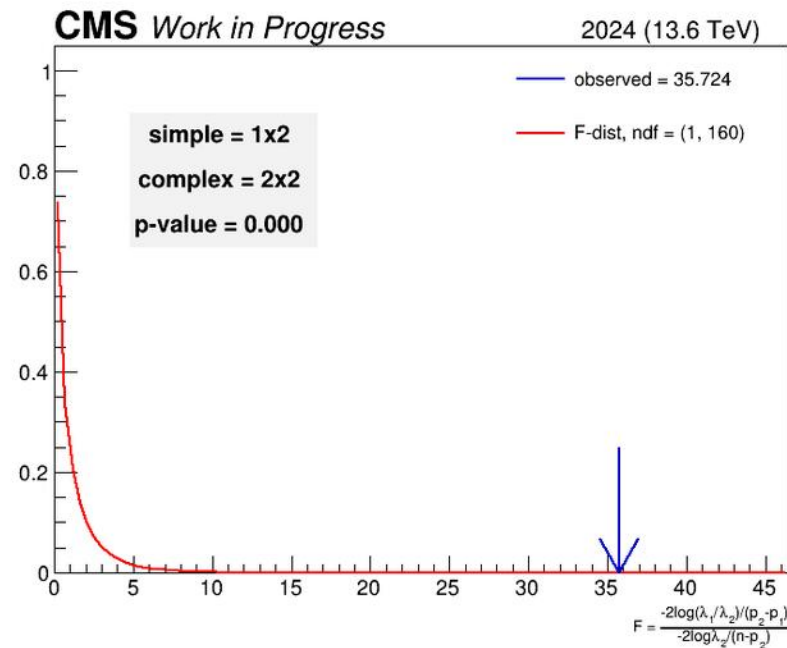
1x1 vs 2x1 | $F = 0$ | $p = 1$



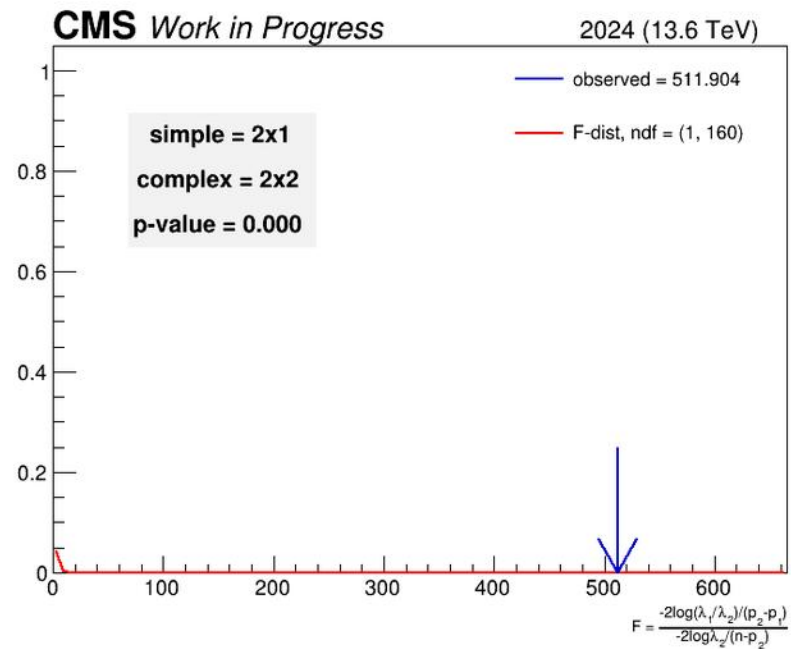
1x1 vs 2x2 | $F = 56.3$ | $p = < \text{numerical precision}$



1x2 vs 2x2 | $F = 35.7$ | $p = 1.43\text{e-}08$



2x1 vs 2x2 | $F = 512$ | $p = < \text{numerical precision}$



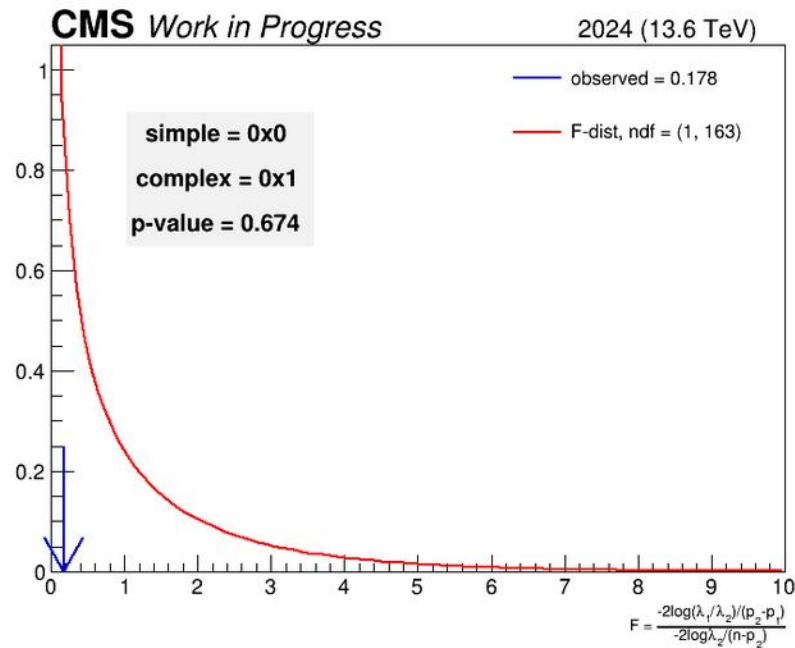
Forward category

25 pairwise comparisons

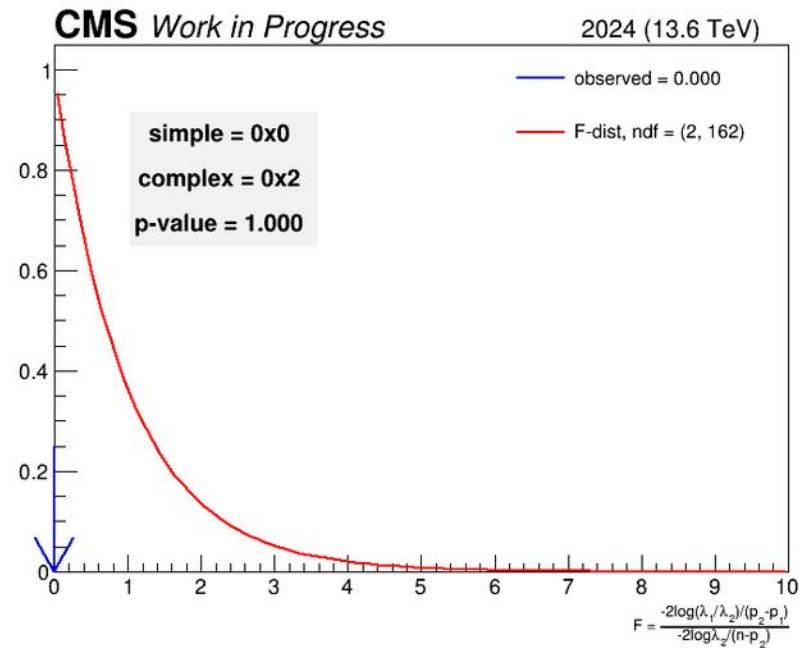
6 comparisons have $p < 0.05$

Counts summarize pairwise tests only; they do not define the final model choice.

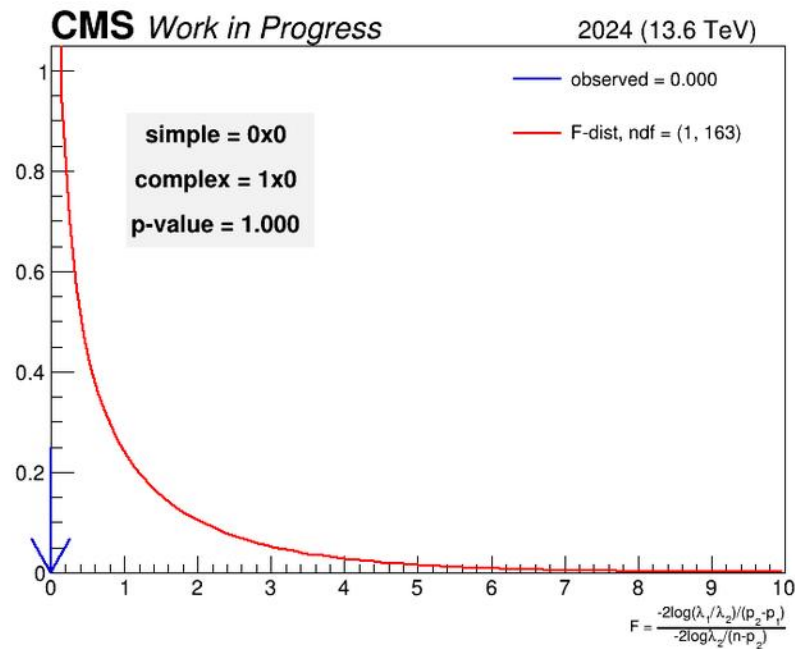
0x0 vs 0x1 | $F = 0.178$ | $p = 0.674$



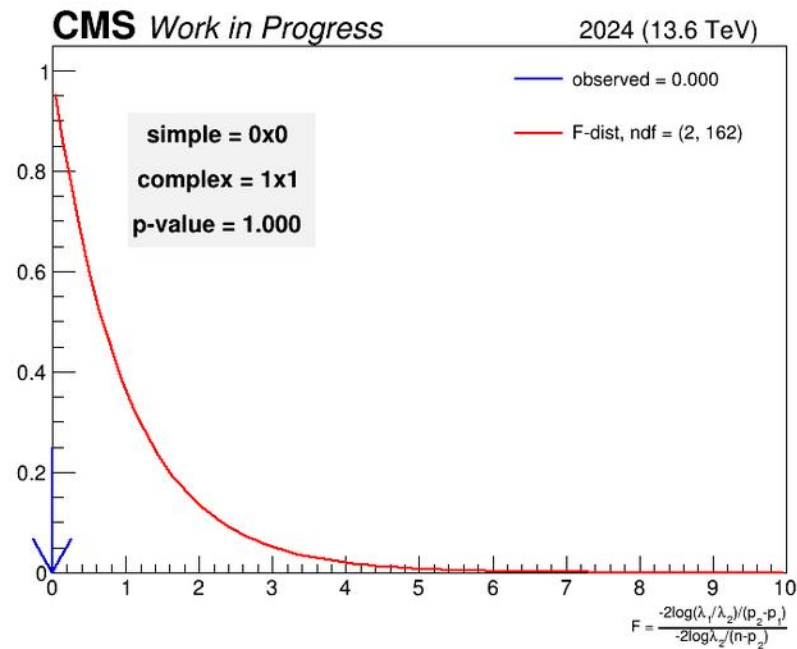
0x0 vs 0x2 | $F = 0$ | $p = 1$



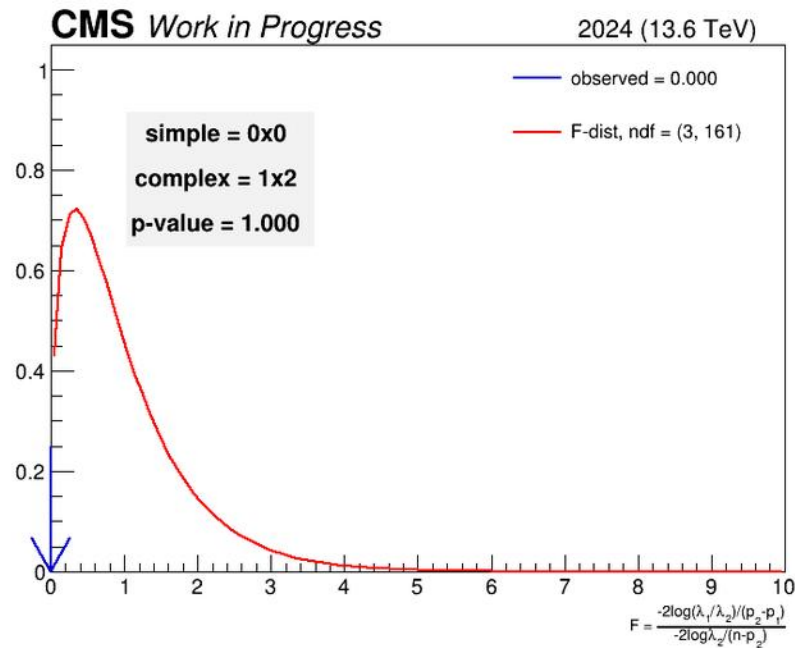
0x0 vs 1x0 | $F = 0$ | $p = 1$



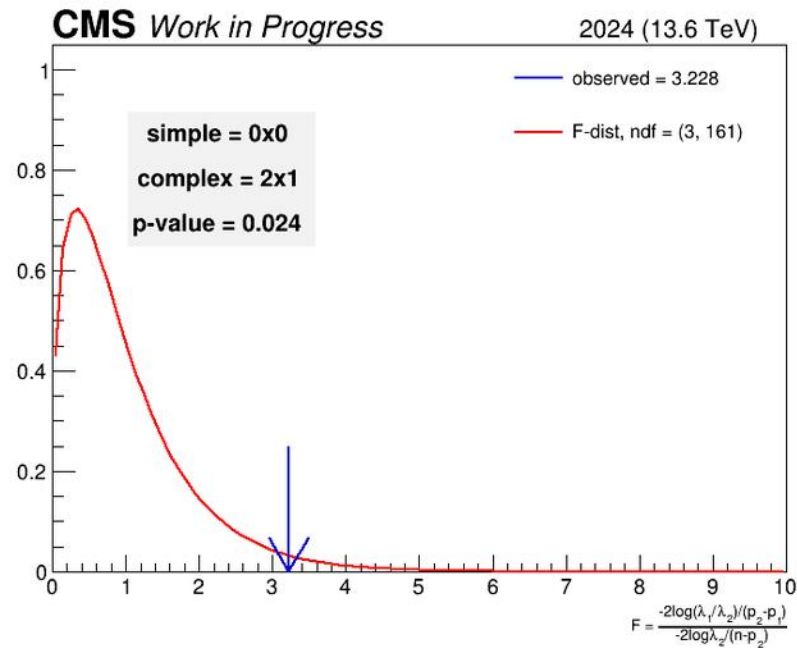
0x0 vs 1x1 | $F = 0$ | $p = 1$



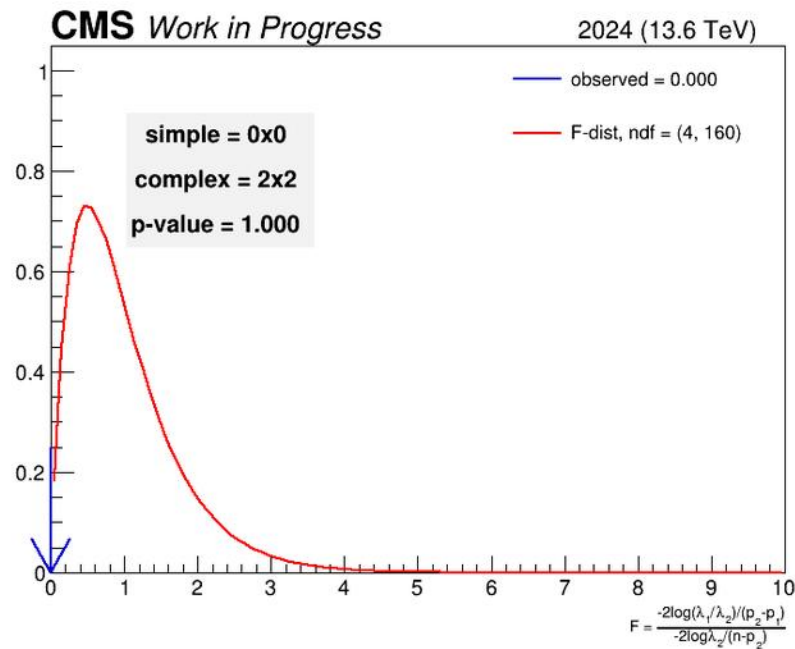
0x0 vs 1x2 | F = 0 | p = 1



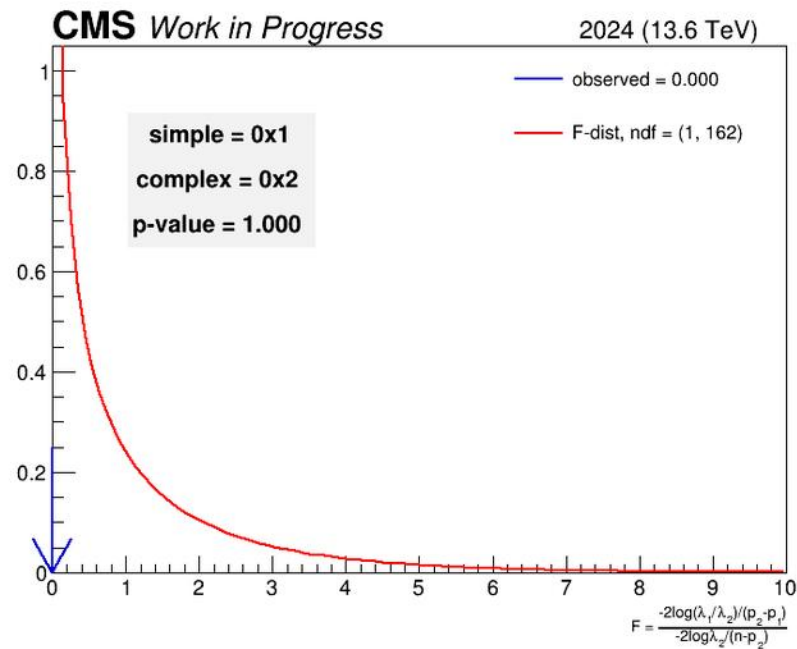
0x0 vs 2x1 | F = 3.23 | p = 0.0241



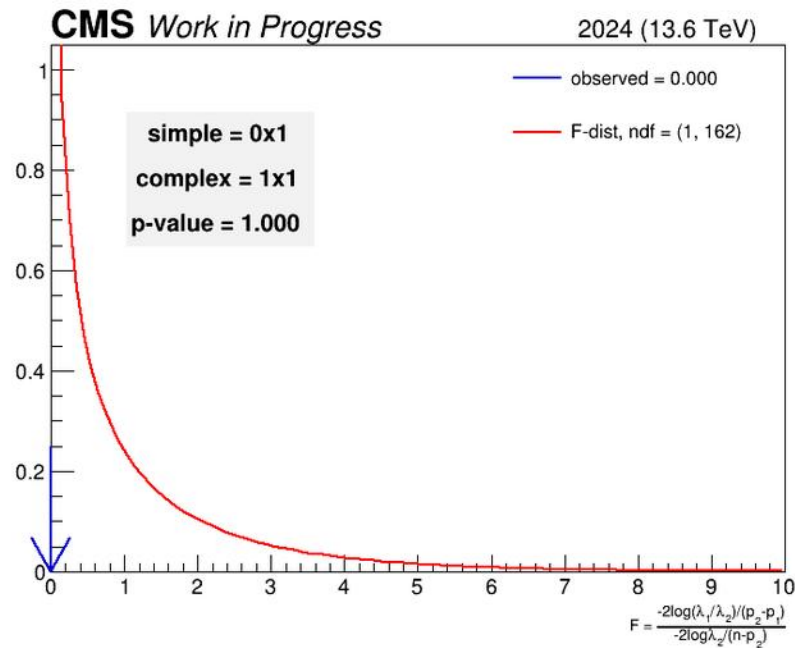
0x0 vs 2x2 | $F = 0$ | $p = 1$



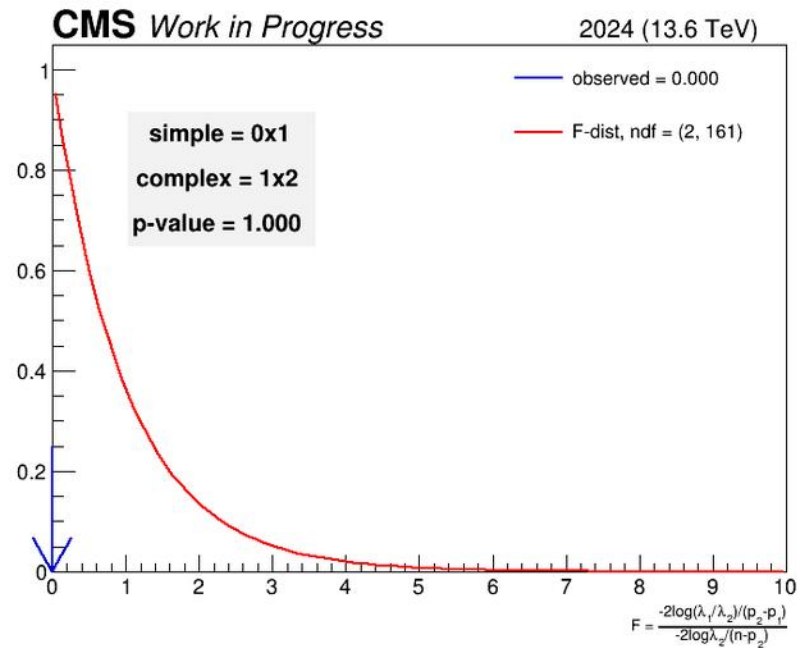
0x1 vs 0x2 | $F = 0$ | $p = 1$



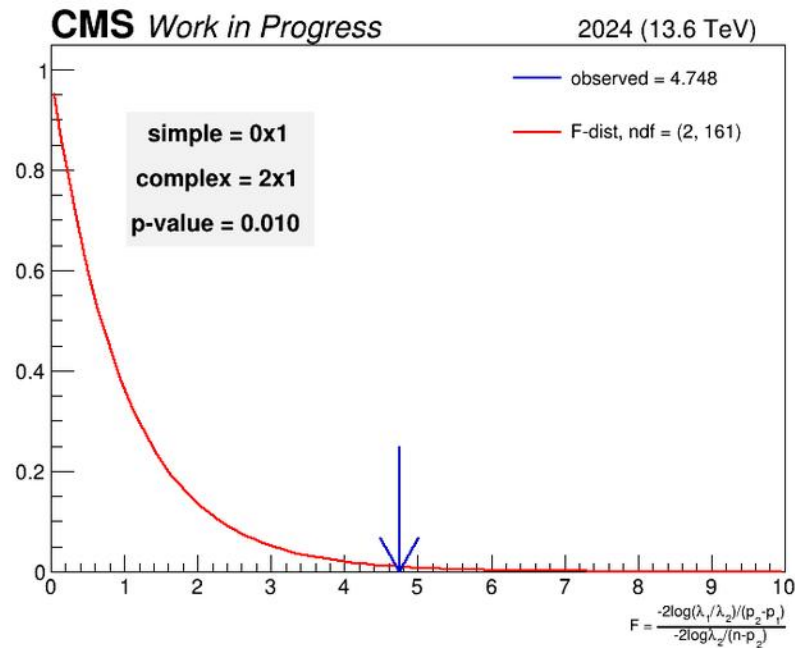
0x1 vs 1x1 | $F = 0$ | $p = 1$



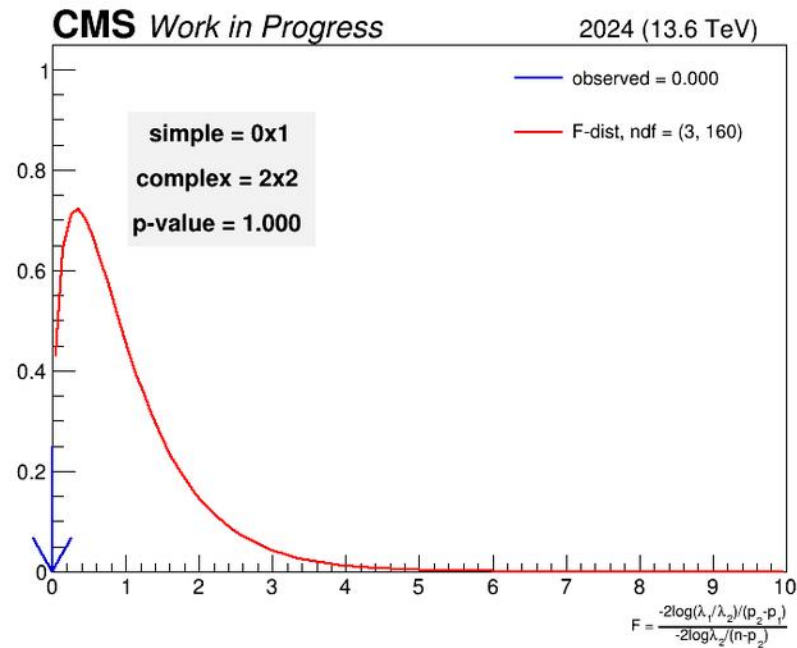
0x1 vs 1x2 | $F = 0$ | $p = 1$



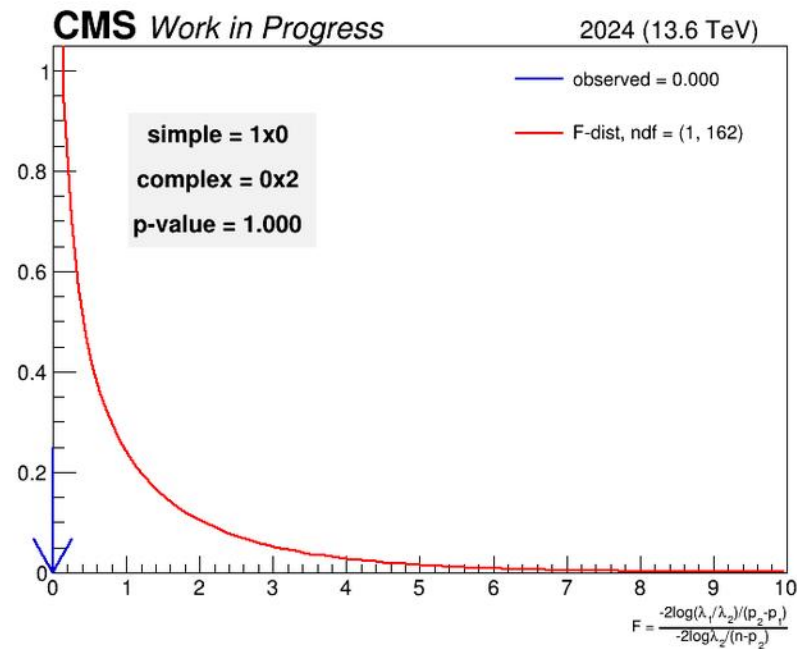
0x1 vs 2x1 | $F = 4.75$ | $p = 0.00991$



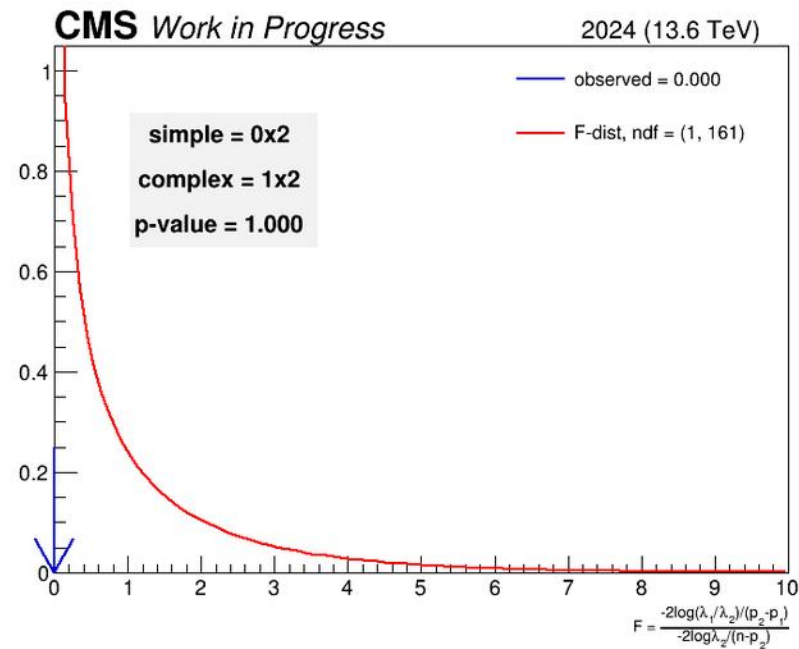
0x1 vs 2x2 | $F = 0$ | $p = 1$



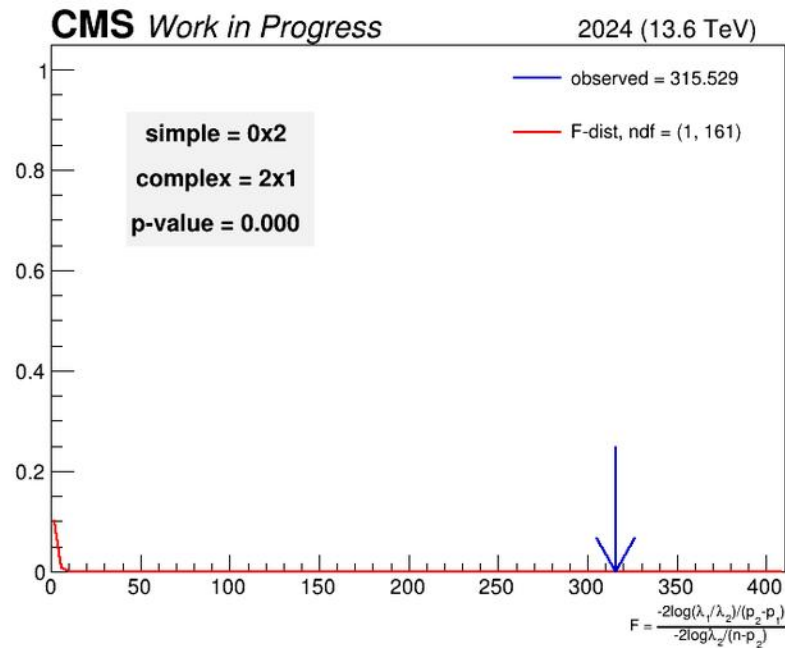
1x0 vs 0x2 | F = 0 | p = 1



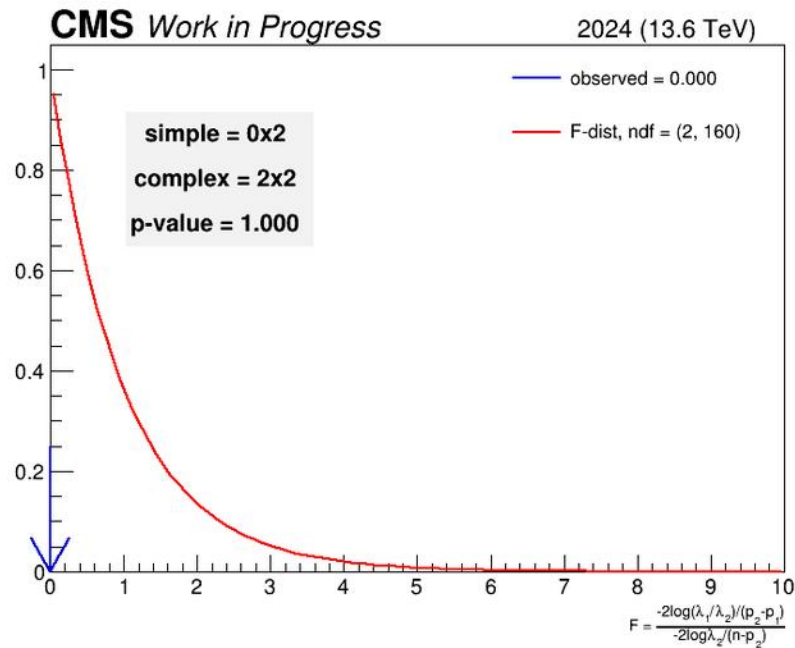
0x2 vs 1x2 | F = 0 | p = 1



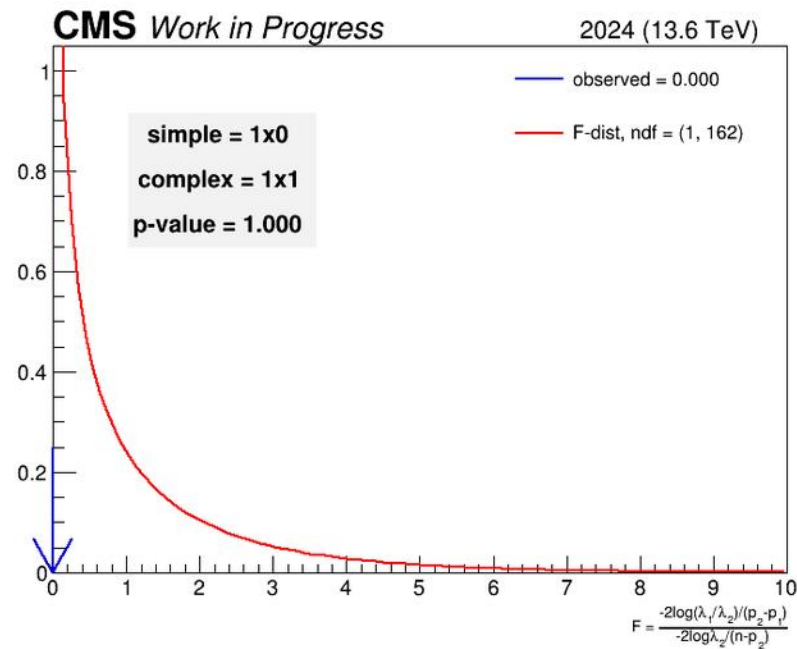
0x2 vs 2x1 | $F = 316$ | $p = < \text{numerical precision}$



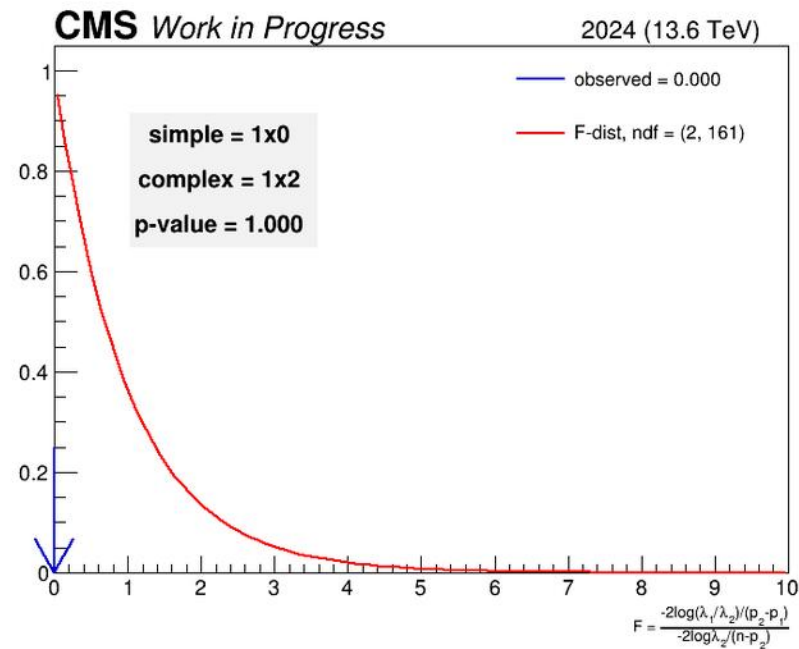
0x2 vs 2x2 | $F = 0$ | $p = 1$



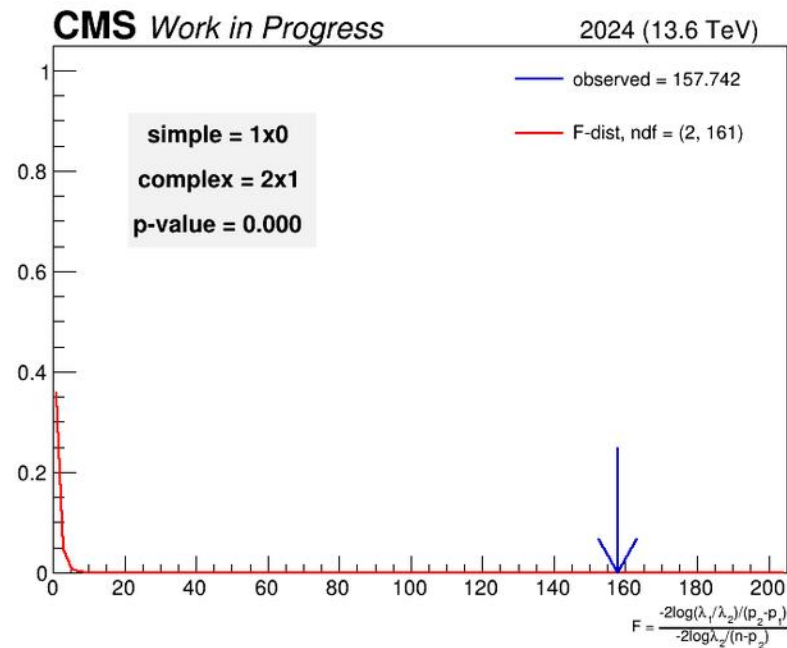
1x0 vs 1x1 | $F = 0$ | $p = 1$



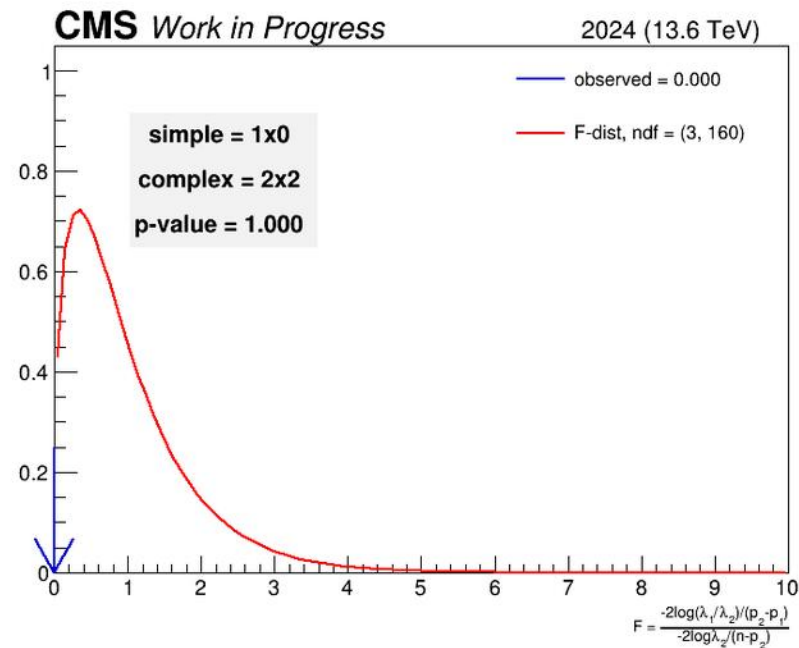
1x0 vs 1x2 | $F = 0$ | $p = 1$



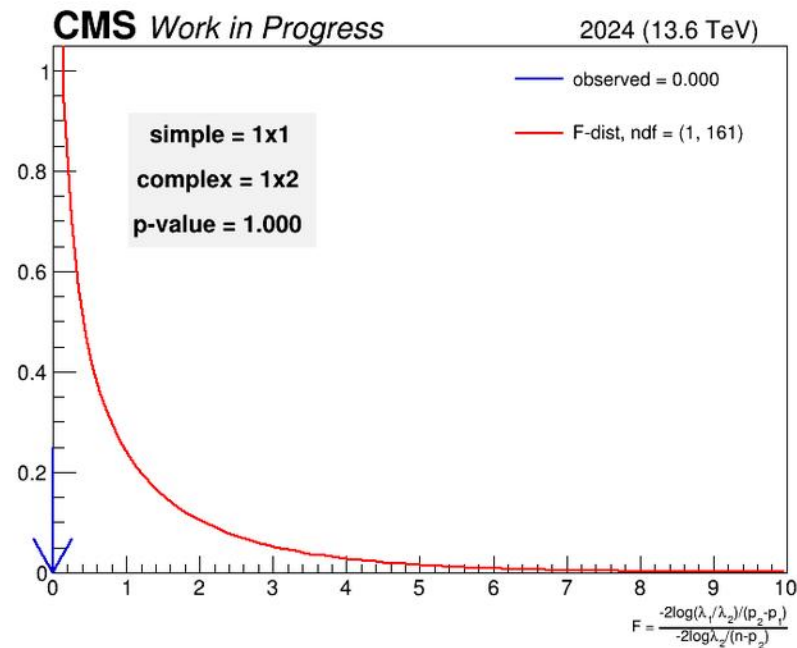
1x0 vs 2x1 | F = 158 | p = 3.7e-14



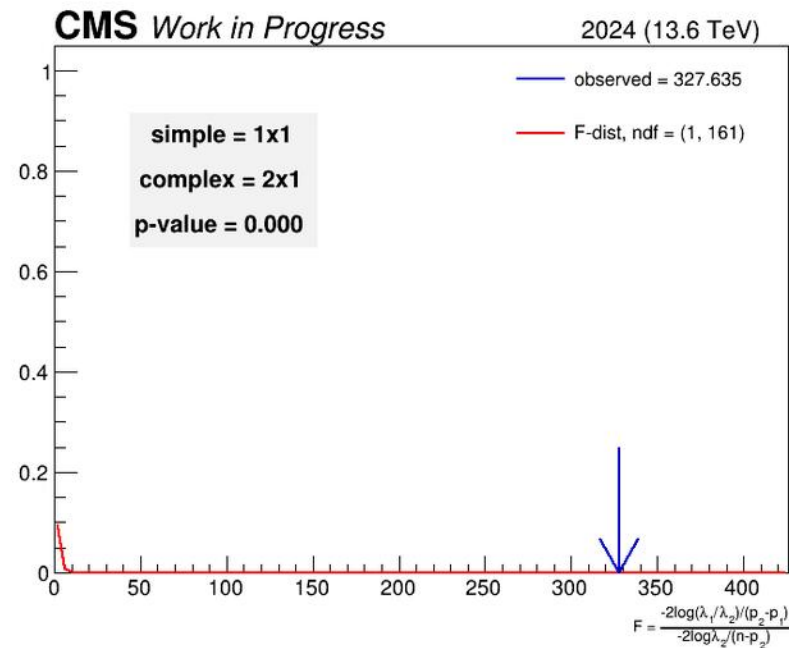
1x0 vs 2x2 | F = 0 | p = 1



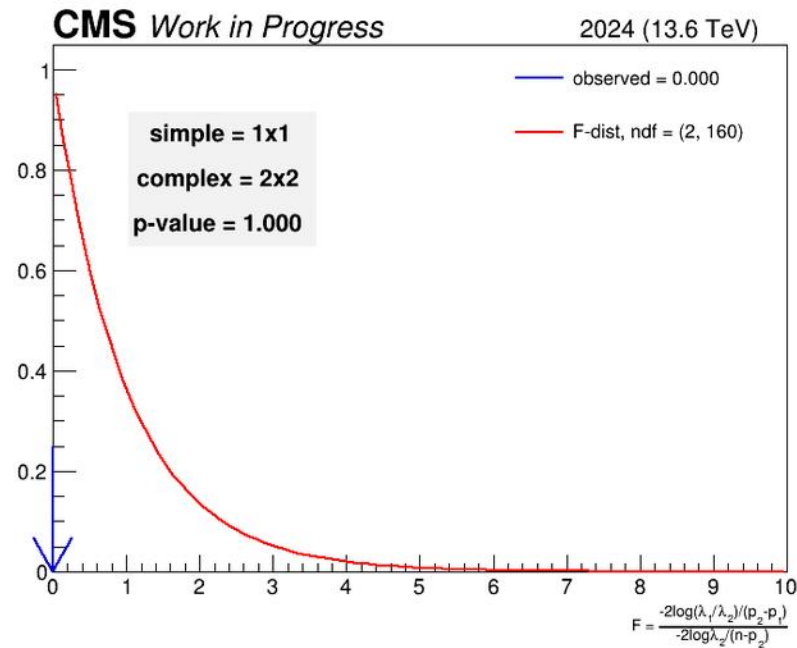
1x1 vs 1x2 | $F = 0$ | $p = 1$



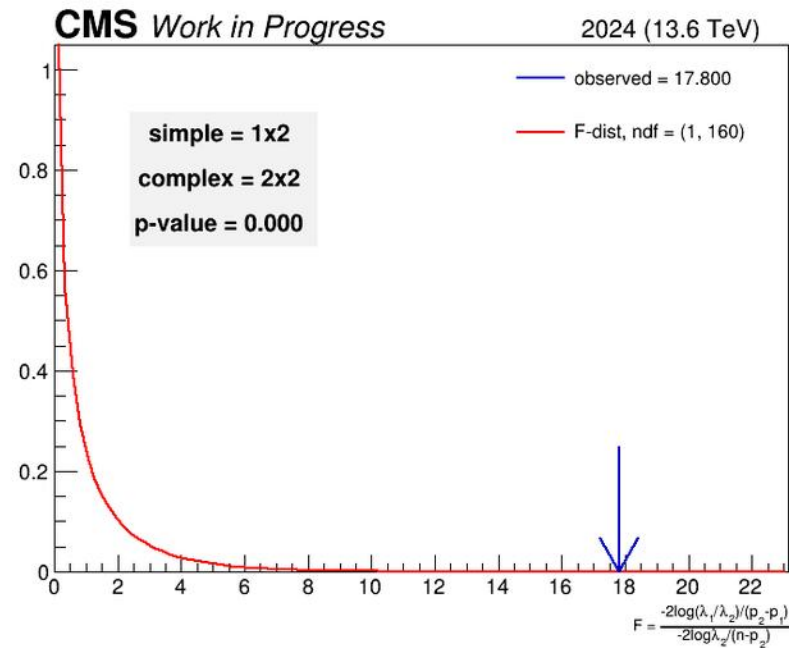
1x1 vs 2x1 | $F = 328$ | $p = < \text{numerical precision}$



1x1 vs 2x2 | $F = 0$ | $p = 1$



1x2 vs 2x2 | $F = 17.8$ | $p = 4.1\text{e-}05$



2x1 vs 2x2 | $F = 0$ | $p = 1$

